

Laheeq Syed

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Profile Summary

Currently employed as a Backend Developer at Tata Consultancy Services (TCS) and presently on a sabbatical leave while pursuing a Master's degree in Computer Science with a specialization in Data Science at Uppsala University. Over the past two years, I have developed and maintained scalable backend services, RESTful APIs, and enterprise applications in Agile environments. Alongside my professional experience, I have expanded my expertise in AI and data-driven systems through advanced research and projects. My current master's thesis focuses on federated learning, parameter-efficient fine-tuning, and large language models, combining scalable software engineering with modern AI technologies. I am passionate about building end-to-end intelligent applications, integrating AI capabilities into software solutions, and collaborating with cross-functional teams to deliver innovative, scalable, and impactful products.

Education

Uppsala University, Sweden

Master's in Computer Science – Computer Science and Engineering (2024–2026) Relevant Coursework: Large Language Models (5/5), Cryptology (4/5), Data Security & Privacy (5/5), Medical Informatics (4/5), Scientific Computation for Data Analysis, Natural Computation for Machine Learning.

R.G.M.C.E.T, India

Bachelor of Technology – Computer Science (2018–2022)

Experience

Systems Engineer, Tata Consultancy Services — Hyderabad, India Jun 2022 – Current

- Developed and maintained backend microservices and data-processing workflows for enterprise scale applications.
- Automated operational and reporting workflows using Java scripting and API integrations, improving internal efficiency and reducing manual overhead.
- Worked with distributed systems, secure data pipelines, and system monitoring in production environments.
- Collaborated across teams to debug, optimize, and deploy scalable software solutions under tight delivery timelines.

Software Developer Intern, Exposys Data Labs — Bangalore, India Jun 2021 – Jul 2021

- Built container level security modules using Docker.
- Gained foundational experience in secure system design.

Relevant Projects

1. MindBridge Agent — AI-Powered Conversational Support System

- Designed and implemented an LLM-powered conversational agent integrating Retrieval-Augmented Generation (RAG) for context-aware response generation.
- Developed prompt engineering and alignment strategies to improve response reliability, safety, and robustness against adversarial or unpredictable user inputs.
- Built workflows for dynamic knowledge retrieval, conversational memory handling, and adaptive response generation using Transformer-based architectures.

2. Furhat ADEM Emotional Recognition & Adaptive Behaviour System

- Developed a multimodal emotional-support agent using Furhat robotics.
- Implemented emotion detection and adaptive response generation.
- Experience with adaptive agents supports GAN-based malicious client design.

3. Social Network Analytics & Data Intelligence Platform

- Processed and analyzed large-scale social network datasets containing over 645K nodes and 39M relationships using Python and NetworkX.

- Built analytical pipelines for user behavior analysis, community detection, and influence discovery across inter-connected datasets.
- Generated insight-driven reports and visualizations from graph-based data to identify high impact users and interaction patterns.

4. Cryptology Research Work — KRACK Attack (WPA2) Course: Cryptology Research

- Analyzed the KRACK (Key Reinstallation) attack exploiting nonce manipulation in WPA2.
- Studied adversarial behavior, protocol vulnerabilities, and mitigation strategies.
- Developed strong understanding of integrity attacks and FL poisoning attacks.

Current Research

Master's Thesis — Efficient and Robust Fine-Tuning for Distributed LLM Systems

- Researching parameter-efficient fine tuning and distributed learning strategies for large language models under resource constrained environments.
- Working with Transformer-based architectures, sparse federated learning frameworks, and scalable experimentation pipelines.
- Focused on improving efficiency, robustness, and adaptability of large-scale AI systems.

Technical Skills

AI & Machine Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Transformer Architectures, Deep Learning, Multimodal Learning, Model Evaluation, Adversarial Robustness

Data & Analytics: Data Analysis, Data Cleaning, Statistical Analysis, Dashboarding, Data Visualization, Business Intelligence, Graph Analytics, Feature Engineering

Frameworks & Libraries: PyTorch, HuggingFace Transformers, Scikit-learn, NetworkX, Pandas, NumPy

Programming: Python, Java, C++, Bash, SQL

Tools & Platforms: Docker, Git, Linux, Jupyter, REST APIs, VS Code, LaTeX

Domains: AI Automation, Data Intelligence, Distributed Systems, NLP, Computer Vision, Secure AI Systems

Data & Analytics: Data Analysis, Dashboard Development, Data Visualization, Business Intelligence, Statistical Analysis, Graph Analytics, Feature Engineering

Frameworks & Libraries: PyTorch, HuggingFace Transformers, Scikit-learn, Pandas, NumPy, NetworkX, Matplotlib

Technologies: Python, PyTorch, HuggingFace, NetworkX, REST APIs

References

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